HOPE LAB / EDUCATIONAL METHOD OVERVIEW

Antimicrobial Susceptibility Testing (AST)

A non-operational overview of what AST measures, why interpretation is method-specific, and how results should be described.

HOPE Lab student reference. This document is maintained as a convenient reference for HOPE Lab students and members. It is not an approved institutional SOP and does not replace supervisor instruction, biosafety review, chemical risk assessment, instrument-specific training, or current institutional procedures. Wet-lab work must be conducted only under an approved, supervisor-directed protocol using appropriately assessed materials and facilities.

Purpose and scope

Antimicrobial susceptibility testing (AST) is a laboratory assessment of how a defined microorganism responds to selected antimicrobial agents under controlled conditions. This document explains the meaning and limitations of AST at an educational level. It does not provide instructions for selecting, handling, preparing, testing, or using antimicrobial drugs or for performing live-organism susceptibility experiments.

What an AST result represents

- An AST result is an in-vitro observation made under a defined laboratory method. It is not, by itself, a treatment recommendation or proof of clinical effectiveness.
- A result is interpretable only in the context of the organism identity, the method used, the agent tested, the applicable validated interpretive framework, and the quality status of the laboratory result.
- Different laboratory approaches may produce different kinds of observations. Results should not be compared or relabelled across methods without an appropriate validated basis.
- Research observations from environmental, food-associated, or incompletely identified isolates should not be presented as clinical susceptibility categories unless the required validated criteria exist and are applicable.

Responsible interpretation and reporting

- Describe the study organism and its identification status at the level actually supported by the evidence.
- State that results are in-vitro observations and identify the general test approach without reproducing operational testing details.
- Report the raw observation and study-specific limitations only when the work was conducted under an approved institutional procedure and interpreted by qualified personnel.
- Do not infer that an organism is clinically treatable or untreatable from an isolated research observation.
- Do not assign susceptible, intermediate, or resistant labels without a validated interpretive basis appropriate to the organism and test context.

Quality and safety boundaries

Practical AST involving antimicrobial agents and live microorganisms belongs in appropriately supervised and authorized laboratories. Method-specific procedures, quality-control limits, interpretive criteria, and reporting decisions must come from the responsible institution and qualified personnel. This document intentionally excludes operational details.

Key takeaway

AST is a specialized laboratory measurement with important method and interpretation constraints. A scientifically responsible report describes the organism, context, observation, method provenance and limitations without overinterpreting the result.

References and authoritative resources

No operational AST method is reproduced in this document. The reference is an educational summary; practical AST methods and interpretive criteria must be obtained and applied by qualified personnel under the relevant institutional quality system.

Document status: literature-informed educational reference. Confirm current editions, organism-specific applicability, and institutional requirements before use. Prepared 10 November 2026.